Negative Gaussian Simplex Stability for Sufficiently Many Parts

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Abstract

We present an argument for the negative Gaussian simplex inequality for every sufficiently large number of parts, throughout the negative voting interval. The comparison permits arbitrary measurable simplex-valued assignments, without prescribed masses, in every finite ambient dimension at least one less than the number of parts. The argument combines competitive mass localization, a deficit-relative local comparison, and an exterior estimate obtained from calibrated covariance derivatives and an exactly centered equal-row test matrix. The local calculation retains a positive centroid square and controls perturbations at all amplitudes; the exterior calculation controls the entire signed Hermite tail. Equality is characterized by regular simplex indicator partitions. Using the Isaksson–Mossel transfer and the credited Frieze–Jerrum endpoint characterization, the conclusion gives negative Plurality Is Stablest and the exact fixed-alphabet MAX-m-CUT approximation threshold, with Unique Games assumed only for hardness. The alphabet threshold is existential. We state all analytic dependencies and provide the supporting proof notes with the source package. The accompanying Lean development formalizes Gaussian routing and integrability, finite covariance identities, scalar asymptotics, and conditional assembly. The comparison estimates themselves remain analytic proof obligations.

Status. This is an AI-assisted research preprint with internal reconstructions of the analytic argument. External expert review and exhaustive priority clearance remain pending. The theorem below is the manuscript's mathematical claim; a successful source build or partial Lean check does not certify its analytic dependencies.

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1 Statement and conventions

Let γ_N be standard Gaussian measure on $\mathbb{R}^N$ and define

$$T_\rho u(x) = \mathbb{E}u(\rho x + \sqrt{1-\rho^2}\, Y), \qquad Y \sim \gamma_N, \quad -1 < \rho < 1.$$

An assignment is a measurable map $f = (f_1, \dots, f_m)$ into $\Delta_m = \{a \in [0,1]^m : \sum_i a_i = 1\}$. Its noise stability is $S_\rho(f) = \sum_i \langle f_i, T_\rho f_i \rangle_{L^2(\gamma_N)}$. This is a single-assignment functional; two independently chosen rules are not permitted in its definition.

Let $P = I_m - \mathbf{1}\mathbf{1}^T/m$. Choose vectors $b_i^0 \in \mathbb{R}^N$ whose Gram matrix is $\alpha^2 P$, where

$$\begin{aligned} &n = m-1, \quad L = 1 + \log m, \quad \mu_m = \mathbb{E}\max_{i \le m} Z_i, \\ &\alpha = \frac{\mu_m}{m-1}, \quad w = \alpha^2(m-1), \quad \beta = \sqrt{w/m}. \end{aligned} \tag{1.1}$$

Here the Z_i are independent standard Gaussians. The regular assignment g chooses the largest $b_i^0 \cdot x$. Ties are Gaussian null. Its centroid matrix is $B_0 = (b_i^0)_i$, and its cells have mass $1/m$. The construction exists for every finite $N \ge m-1$ and extends cylindrically when $N > m-1$.

Theorem 1.1. *There is an absolute integer m_0 such that, for every $m \ge m_0$, every finite $N \ge m-1$, every $\rho \in [-1/(m-1), 0)$, and every measurable assignment $f : \mathbb{R}^N \to \Delta_m$,*

$$S_\rho(f) \ge S_\rho(g).$$

Equality holds precisely for the indicator assignments of regular Gaussian simplex partitions, up to orthogonal transformations, relabeling, and Gaussian null sets. No mass constraint is imposed on f. The threshold m_0 is independent of ρ and of finite N.

This manuscript provides no numerical value of m_0. It does not claim the positive-correlation conjecture, the remaining small values of m, all negative Gaussian correlations, or a bilinear theorem. Section 10 gives the exact negative Plurality is Stablest consequence and the UGC-conditional Frieze–Jerrum threshold. These are fixed-m consequences, with the endpoint formula credited to García-Soriano–Miyauchi [37].

The original negative Standard Simplex formulation has no mass constraint [19, Conjecture 1.4]. The older large-m endpoint estimate in [20, Proposition 11.4], recalled in [19, equation (20)], is asymptotic, with an additive error of order $\log\log(m)/m^2$. Theorem 1.1 instead asserts exact optimality for each fixed sufficiently large m on an entire interval. The bounded primary-source check [18] is not an exhaustive priority or novelty determination.

For an assignment write

$$p_i = \mathbb{E} f_i, \qquad B_i = \mathbb{E}[X f_i], \qquad W = \|B\|_{\mathrm{F}}^2, \qquad U = m^{-1}\mathbf{1}. \tag{1.2}$$

Always $B^T\mathbf{1} = 0$. For a centered score matrix B define

$$F_p(B) = \sup\left\{ \mathbb{E}\sum_i f_i B_i \cdot X : f : \mathbb{R}^N \to \Delta_m,\ \mathbb{E} f = p \right\}, \tag{1.3}$$

$$\mathcal{R}(B) = \frac{w + \|B\|_{\mathrm{F}}^2}{2} - F_U(B). \tag{1.4}$$

Its value depends only on BB^T. Let $\Gamma(C) = F_U(C^{1/2})$ for a centered positive semidefinite matrix C; all square-root realizations are equivalent by Gaussian invariance. The regular orbit distance is $d(B) = \inf_{B_0}\|B - B_0\|_{\mathrm{F}}$, over regular centroid matrices in the same ambient space. Constants denoted c, C may change from line to line. A subscript records fixed parameter dependence. All estimates called uniform below are independent of finite N and of the correlation in the stated interval.

2 Analytic interfaces and dependency structure

The following interfaces record the hypotheses used by the final assembly. They are mathematical proof obligations, not consequences of an audit label. Sections 3, 8, and 7 reconstruct their principal arguments; the accompanying analytic notes listed in Appendix A supply the longer finite-fan, transport, and boundary calculations. Those notes form part of the proof package and are required to assess the complete claim. The external Gaussian comparison is separately identified below. The radial estimate in Section 5 is not assumed as an interface.

Analytic interface 2.1 (Covariance value, normal strain, and offsets)**.** *The functional Γ is concave and continuous on centered positive semidefinite matrices. At a positive definite matrix on $\mathbf{1}^\perp$, its derivative is $\mathcal{L}/2$, where $\mathcal{L}$ is the mass-offset Laplacian. In particular $\Gamma(P) = \mu_m$. It is C^2 locally in the positive cone. At $C_0 = \alpha^2 P$, the normal-strain estimate implies*

$$D^2\Gamma(C_0)[E,E] < 0 \quad \text{if some } (e_i - e_j)^T E (e_i - e_j) \neq 0, \tag{2.1}$$

for a centered symmetric E. Every regular facet has positive Gaussian measure, so this strictness criterion applies to each pair.

If $\|B\|_{\mathrm{F}}^2 = w$, $\max_i |B_i| \le A\alpha$, and h_U is a mean-zero balanced optimal offset, then

$$\|h_U\|^2 \le C_A L^2 \mathcal{R}(B). \tag{2.2}$$

This is a value/offset statement at arbitrary rank, including coincident sites by continuity. Positive scaling gives

$$\mathcal{R}(sB) = \tfrac{1}{2}w(s-1)^2 + s\mathcal{R}(B), \qquad h_U(sB) = sh_U(B) \quad (s > 0,\ \|B\|^2 = w). \tag{2.3}$$

In the mass box $p_i \in [1/(2m), 3/(2m)]$, the optimum first-moment value $V(p) = \sup_{\mathbb{E}f=p}\|M_f\|^2$ satisfies $V(p) \le w + CL^2\|p-U\|^2$.

The covariance and normal-strain input is supplied by the classical-input Hodge argument [27], summarized in Section 3. The offset and mass-envelope argument additionally uses the stronger cycle-refined Ehrhard inequality [28, 29]; this dependency is not removed by the Hodge replacement.

Proposition 2.2 (Endpoint and zero-remainder rigidity)**.** *For every centered B, $\mathcal{R}(B) \ge 0$, with equality precisely when $BB^T = \alpha^2 P$. Every balanced measurable assignment has $W \le w$; equality forces the regular assignment. More generally, if an arbitrary assignment's actual centroids have $\mathcal{R}(B) = 0$, it is regular.*

Proof. We give the direct covariance proof [21]. For iid Gaussian coordinates, the expected winning coordinate is μ_m. Label symmetry gives diagonal centroid entry μ_m/m. The common coordinate mean is independent of the winning label and has mean zero. The off-diagonal entries are equal by symmetry, and centering makes the centroid matrix exactly $B_0 = \alpha P$ on $\mathbf{1}^\perp$. Its masses are $1/m$ and $F_U(B_0) = \alpha\mu_m = w$.

The reference winner is unique off Gaussian-null hyperplanes. Hence the score-value envelope is differentiable there and has derivative B_0. Differentiating $C = BB^T$ at B_0, which is invertible on $\mathbf{1}^\perp$, gives $\nabla\Gamma(C_0) = P/2$. Concavity therefore yields

$$\Gamma(C) \le w + \tfrac{1}{2}(\operatorname{tr} C - w)$$

for every centered positive semidefinite C, proving nonnegativity. If $C \ne C_0$, put $E = C - C_0$. If every edge quadratic in (2.1) vanished, writing $d_i = E_{ii}$ would give $E_{ij} = (d_i + d_j)/2$. The row sums then imply $md_i + \operatorname{tr} E = 0$ for each i, and summing forces $\operatorname{tr} E = 0$ and $E = 0$, a contradiction. Thus the path $C_0 + tE$ has strictly negative second derivative at zero, and by local C^2 regularity on a short initial interval. Its positive gap below the reference tangent is thereafter nondecreasing by concavity. Continuity handles a singular endpoint at $t = 1$. This proves $\mathcal{R}(B) > 0$ unless $BB^T = C_0$.

For balanced f with actual centroids B, feasibility gives $W \le F_U(B) \le (w + W)/2$, hence $W \le w$. Equality implies $BB^T = C_0$. For any assignment with this actual Gram matrix, $W = w = \mathbb{E}\max_i B_i \cdot X$. Its nonnegative routing loss $\max_i B_i \cdot X - \sum_i f_i B_i \cdot X$ has expectation zero. The regular rows have Gaussian-null ties, so f is their regular winning assignment. This final implication needs no mass assumption. □

Analytic interface 2.3 (Competitive localization)**.** *For all sufficiently large m and $0 < r \le 1/(m-1)$, an assignment with $S_{-r}(f) \le S_{-r}(g)$ has $p_i \in [1/(2m), 3/(2m)]$. Put*

$$\delta = \|p - U\|^2, \quad \Delta = w - W, \quad D = \delta + r\Delta.$$

Then

$$0 \le \delta \le 2D, \qquad 0 \le D \le Cr^2/m. \tag{2.4}$$

The proof is reconstructed in Section 3 from [5, 29]; it applies to arbitrary competitive assignments. For every assignment, conditional Cauchy gives $(B_i \cdot v)^2 \le p_i \mathbb{E}[f_i(X)(v \cdot X)^2]$. Summation shows

$$B^T B \preceq p_{\max} I. \tag{2.5}$$

Conditional Jensen in $\mathbb{E}e^{tv\cdot X}$ similarly gives

$$|B_i| \le p_i \sqrt{2\log(1/p_i)}. \tag{2.6}$$

Neither estimate requires stationarity or purity.

Analytic interface 2.4 (Global one-row covariance coercivity)**.** *For every fixed A, every centered score matrix with $\|B\|^2 = w$ and $\max_i |B_i| \le A\alpha$ satisfies*

$$\max_i \left| \frac{|B_i|^2}{\beta^2} - 1 \right|^2 \le C_A \frac{m^2}{L} \mathcal{R}(B). \tag{2.7}$$

No rank, assignment-realizability, or minimization hypothesis is imposed.

This is the global single-row estimate of [6, Theorem 1]. It will be applied directly to a normalized actual centroid matrix; no actual-minimizer row theorem is assumed.

Analytic interface 2.5 (Global covariance remainder and local full-noise comparison)**.** *For centered B, put $C = BB^T/\alpha^2$ and $e = \|C - P\|_{\mathrm{F}}$. The initial-segment covariance bound is*

$$\mathcal{R}(B) \ge \frac{c\alpha^2 e^2}{m} \min\left\{1, \frac{c}{Le}\right\}. \tag{2.8}$$

It holds at arbitrary endpoint rank and does not require $\|B\|^2 = w$; the value at $e = 0$ is interpreted by continuity.

For every fixed A, sufficiently large m, and every measurable assignment whose actual *centroids obey $d(B) \le A/m$ and whose masses lie in $[1/(2m), 3/(2m)]$,*

$$S_{-r}(f) - S_{-r}(g) \ge \tfrac{1}{2}\|p - U\|^2 + r\mathcal{R}(B), \qquad 0 < r \le 1/(m-1). \tag{2.9}$$

Competitiveness in this statement forces a regular assignment.

Section 7 proves the global ray estimate from the quantitative Hodge input and the facet bootstrap [30]. Section 8 reconstructs the local theorem [31]. Its complete cap and affine-boundary calculations are in the accompanying source notes. It concerns arbitrary measurable assignments, including fractional rules.

Analytic interface 2.6 (Regular normalization and margin)**.** *Let Π_k denote the orthogonal Hermite projection of degree k and $A_k(f) = \sum_i \|\Pi_k f_i\|_2^2$. Then*

$$A_2(g) = \frac{m\{\mathbb{E}[(\max_i Z_i)^2] - 1\}^2}{2(m-1)(m-2)}, \qquad \alpha^2 \asymp L/m^2. \tag{2.10}$$

For a regular score matrix B_0, let $G(X)$ be its largest score minus its second largest score. Then

$$\mathbb{P}\{G \le t\} \le Cmt \quad (t \ge 0). \tag{2.11}$$

The exact second-chaos identity is in [5]; the margin and its fractional routing consequence are in [10]. We also use the standard dimension-free Gaussian hypercontractivity $\|T_t u\|_2 \le \|u\|_{1+t^2}$ [26], and the one-dimensional variance bound $\mathrm{Var}_\nu x \le 1/c$ when the density of ν has logarithmic second derivative at most $-c$ [24]. The transitive proofs of the offset and mass-window inputs also use Gaussian isoperimetry [25] and the convex Ehrhard inequality [23]; their precise uses are recorded in the companion section-level source map.

Analytic interface 2.7 (Equal-threshold Gaussian maximum comparison)**.** *If K is an $m \times m$ correlation matrix and $K - \mathbf{1}\mathbf{1}^T/m \succeq 0$, then for $Y \sim N(0,K)$ and independent standard Gaussians Z_i,*

$$\mathbb{P}\{\max_i Y_i \le t\} \ge \Phi(t)^m \qquad (t \in \mathbb{R}).$$

The statement includes singular K.

This is Theorem 2.1 of the pinned primary source [15]. Only equal thresholds are used. We do not infer a general rectangular comparison or a comparison for unequal marginal variances.

3 Covariance foundation and competitive localization

This section reconstructs the covariance, mass, and localization inputs. The Hodge proof and the cycle refinement are distinct components. The former gives covariance concavity and normal strain; the latter is needed for the mixed covariance/offset estimate in the mass envelope. Detailed weak-limit and finite-fan identities are supplied in the analytic notes [27, 28, 29].

3.1 The covariance functional and its quantitative curvature

For a probability law ν on $\mathbb{R}^d$ with finite second moment, set

$$\mathcal{F}_\nu(C) = \sup\{\mathbb{E}X \cdot Y : X \sim N(0,C),\ Y \sim \nu\}.$$

The classical-input Hodge argument [27] establishes concavity on the ordinary PSD covariance cone, including singular covariances by continuity. It does not concern Bures geodesics. For a simplex target $\nu = \sum_{i=1}^{d+1} p_i \delta_{y_i}$ with affinely independent sites and positive masses, write $a_{ij} = y_i - y_j$ and let $s_{ij}(C)$ be the Gaussian surface area of the physical interface for the scores $C^{1/2}y_i$ with the prescribed-mass offsets. Put $w_{ij} = s_{ij}/|C^{1/2}a_{ij}|$. Then

$$D\mathcal{F}_\nu(C)[H] = \frac{1}{2}\sum_{i<j} w_{ij} a_{ij}^T H a_{ij}, \tag{3.1}$$

$$-4D^2\mathcal{F}_\nu(C)[H,H] \ge \sum_{i<j} w_{ij}\frac{(a_{ij}^T H a_{ij})^2}{a_{ij}^T C a_{ij}}. \tag{3.2}$$

Strictness holds on the affine span of the simplex. The balanced target $y_i = Pe_i$ gives the functional Γ used here.

We record the route to these inequalities to identify the substantial analytic input. If $M = D^2\varphi$ is smooth and uniformly elliptic, the weighted Gaussian Hodge estimate gives, for every weak Gaussian-solenoidal vector field v, a skew matrix field A such that

$$\delta_\gamma A = v, \qquad \mathbb{E}\,\mathrm{tr}(A^T M A) \le \mathbb{E}v^T M v.$$

Its proof minimizes the left side subject to the divergence constraint. The Euler equation is $d(MA+AM)=0$. The integration-by-parts identity reduces its three-index terms to

$$-2(\beta ab+\gamma bc+\alpha ca)+\frac{[(\alpha+\beta)a+(\beta+\gamma)b+(\gamma+\alpha)c]^2}{2(\alpha+\beta+\gamma)}\geq 0$$

for positive eigenvalues α,β,γ. The remaining terms are nonnegative squares. Gaussian OU resolvents, compact cutoffs, and graph approximation justify the weak minimization and integration by parts; these steps are proved in the supplied Hodge notes, not by the finite identity alone.

For symmetric H define

$$j_M(H)=\sum_i \lambda_i H_{ii}^2+\sum_{i<j}\frac{4\lambda_i\lambda_j}{\lambda_i+\lambda_j}H_{ij}^2,$$

in an eigenbasis of M. The coupling variational formula and the Hodge bound give the local upper Peano estimate

$$\limsup_{t\to 0}\frac{\mathcal{F}_\nu(I+tH)-\mathcal{F}_\nu(I)-(t/2)\operatorname{tr}(H\mathbb{E}M)}{t^2}\leq -\frac{1}{8}\mathbb{E}j_M(H). \tag{3.3}$$

Strong convexity bounds optimizing displacements in L^2 by $O(t)$; Fenchel duality supplies the energy lower limit for their weak velocities. This avoids differentiating an optimal transport map along the covariance path or assuming a uniform Taylor remainder for unbounded displacements.

The smooth target class $Z^{-1}e^{-q}\gamma_\Sigma$, with compactly supported smooth q, is covered by Colombo–Figalli–Jhaveri [39]. The reciprocal change $x'=\Sigma^{1/4}x$, $y'=\Sigma^{-1/4}y$ makes the two reference Gaussians identical and preserves the coupling inner product. The class is closed under invertible linear target changes and is dense in W_2. A scalar chord-and-quadratic barrier applied to (3.3) gives concavity along each covariance segment. The bound

$$|\mathcal{F}_\nu(C)-\mathcal{F}_\eta(C)|\leq\sqrt{\operatorname{tr}C}\,W_2(\nu,\eta)$$

passes it to arbitrary targets; Gaussian W_2 continuity includes singular source covariances.

For simplex targets, smooth Hessians converge as PSD matrix measures to rank-one facet measures. The representation

$$j_M(H)=\min_{K^T=-K}\operatorname{tr}((H+K)^TM(H+K))$$

and the estimate, for a unit vector n,

$$|j_M(H)-\operatorname{tr}(M)(n^THn)^2|\leq 16\|H\|_{\mathrm{F}}^2\sqrt{\operatorname{tr}(M)\operatorname{tr}((I-nn^T)M)}$$

give the required facet limit. This is a rank-one measure argument, not a general lower-semicontinuity claim for a concave matrix function. Patch exhaustion and a scalar curvature barrier yield (3.2). The Gaussian heat identity and the offset envelope give (3.1), including its factor $1/2$.

3.2 The mass envelope and the retained cycle refinement

Fix absolute $0<c<1<\overline{C}$ and let $c/m\leq p_i\leq\overline{C}/m$. Square completion gives

$$V(p)=\sup_{\mathbb{E}f=p}\|M_f\|_{\mathrm{F}}^2=\max_{C\succeq 0}\{2\Gamma_p(C)-\operatorname{tr}C\}, \tag{3.4}$$

where $\Gamma_p(C)$ is the Gaussian linear assignment value with prescribed masses. In this mass-envelope argument, covariance and mixed derivatives written as Γ_{CC}, Γ_{Cp}, or Γ_{pp} are derivatives of $\Gamma_p(C)$ at the current prescribed mass p; they are not restricted to the balanced value $\Gamma = \Gamma_U$. Its support function agrees for pure and fractional rules. A pure support maximizer exists by the finite Gaussian offset dual; duplicate-score ties may be split measurably. This does not assert purity of a full-noise minimizer.

Restrict the covariance in (3.4) to $C \succeq \epsilon I_H$ on $H = \mathbf{1}^\perp$, and write the resulting value as V_ϵ. The elementary bound $\Gamma_p(C) \le \sqrt{\operatorname{tr} C}$ gives a common trace bound. Adding an independent Gaussian to a coupling gives

$$0 \le V(p) - V_\epsilon(p) \le \epsilon(m-1). \tag{3.5}$$

Indeed, if (X, Y) is an admissible coupling for $\Gamma_p(C)$, retain Y and replace X by $X + G$, where G is an independent centered Gaussian of covariance ϵI_H. The objective is unchanged, while the source covariance becomes $C + \epsilon I_H$. Thus $\Gamma_p(C + \epsilon I_H) \ge \Gamma_p(C)$, including singular C; the trace penalty increases by exactly $\epsilon(m-1)$. At a regularized optimizer, the derivative formula and KKT conditions give

$$0 \preceq \mathcal{L} \preceq I_H, \qquad (I_H - \mathcal{L})(C_\epsilon - \epsilon I_H) = 0.$$

Thus physical graph degrees are bounded by one. A two-cell re-splitting argument at an unregularized optimum, using the scalar Gaussian density bound, gives $|B_i - B_j| \ge \sqrt{2\pi} p_i p_j$. The entropy estimate gives the upper edge-length bound $C\sqrt{L}/m$. Compactness of the closed optimizer graph transfers these inequalities with slack to the regularized optimizers; continuity of individual optimizer fibers is unnecessary.

Gaussian isoperimetry applied to unions of cells gives conductance at least $c_1/\sqrt{L}$. Removing edges of weight below m^{-4} loses at most $|S|m^{-3}$ across a cut. The retained graph therefore has counting-measure gap $\lambda_* \ge c_2/L$. Its intrinsic facet probabilities satisfy $\pi_e \ge c_3 m^{-6}$, and the normalized offsets satisfy $q_e^2 \le 2\log(m/c) \le C_c L$, where $p_i \ge c/m$ is the fixed lower mass bound. Constants in this subsection may depend on the fixed mass-box parameters c and $\overline{C}$, but not on dimension or noise. These are facet probabilities, not the masses of the adjoining cells.

The stronger finite-fan estimate used here is

$$-4\Gamma_{CC}[A, A] \ge \sum_e w_e\{x_e^2 + a(\pi_e)(2h'_e - q_e x_e)^2\}, \qquad a(p) = 1 + \frac{\Phi^{-1}(p)p}{\phi(\Phi^{-1}(p))}. \tag{3.6}$$

Here $d_e = y_i - y_j$ is an edge at the whitened covariance base, $\widetilde{A}$ is the whitened covariance perturbation, $x_e = d_e^T \widetilde{A} d_e/|d_e|$, and $q_e = (h_i - h_j)/|d_e|$. The vector h' is the mass-balancing offset derivative, with $h'_e = h'_i - h'_j$. The proof in [28] retains the whole cycle correction before applying intrinsic Ehrhard. More precisely, the finite-fan integration identities yield a negative semidefinite weighted boundary-of-boundary remainder. Choosing its weighted gauge preserves the prescribed cycle j. If r_e is the resulting residual tangential field and $E_e = w_e(P_e - \operatorname{Cov}_e X)$, the two identities used are

$$\sum_e r_e^T E_e r_e = \sum_e z_e^T E_e z_e - \mathrm{CYC}, \qquad r_e \cdot P_e m_e = -\mathrm{d}b_e - z_e \cdot (I - P_e)m_e.$$

Intrinsic Ehrhard is then applied to r_e, with $b = 2h'$. Applying it directly to z_e would discard the cycle energy and would not prove (3.6). The complete finite-fan definitions and identity proof are in [28, Sections 3–5]; the current reconstruction is [29, Section 2b].

On retained edges, intrinsic Ehrhard gives $a(\pi_e)^{-1} \le CL$. The elementary inequality

$$4(h'_e)^2 \le (a(\pi_e)^{-1} + q_e^2)\{x_e^2 + a(\pi_e)(2h'_e - q_e x_e)^2\}$$

and the graph gap consequently imply

$$\|D_C h[A]\|^2 \le CL^2[-\Gamma_{CC}[A,A]]. \tag{3.7}$$

The normal-strain estimate alone would not imply this mixed bound.

Let $g(C,p) = 2\Gamma_p(C) - \operatorname{tr} C$. Along a mass segment $p(t)$ with velocity u, optimality inequalities at two times imply almost everywhere

$$E_0 = -g_{CC}[C',C'] \le B_0 = g_{Cp}[C',u], \qquad B_0^2 \le CL^2|u|^2 E_0.$$

For fixed ϵ, strict curvature and compactness give a Lipschitz optimizer map and a Lipschitz envelope derivative. Since $\Gamma_{pp} = -\mathcal{L}^{-1}$, the second derivative of the envelope is at most $CL^2|u|^2$. At U, symmetry gives zero centered derivative. Integrating, passing $\epsilon \downarrow 0$ in (3.5), and using $V(U) = w$ prove

$$V(p) \le w + CL^2|p-U|^2. \tag{3.8}$$

The regularization cutoff may depend on fixed m; the coefficient in (3.8) and the eventual noise threshold do not.

3.3 From arbitrary competitive rules to a narrow mass box

Suppose $S_{-r}(f) \le S_{-r}(g)$. Scalar Gaussian rearrangement gives $|B_i|^2 \le H(p_i)$, where $H(p) = \phi(\Phi^{-1}(p))^2$ with continuous endpoint values. The global scalar tangent estimate is

$$\sum_i H(p_i) \le mH(1/m) + B_m\delta, \qquad B_m = O(\log m).$$

Hermite parity gives

$$S_{-r}(f) \ge \|p\|^2 - rW - r^3, \qquad S_{-r}(g) \le m^{-1} - rw + r^2 A_2(g) + r^4.$$

The Mills/Gumbel calculation in [5], including uniform integrability of the first two centered maximum moments, yields $m^2\delta \le L_m$ with $L_m \to 2 - 2\gamma_{\mathrm{E}} < 1$. Consequently a fixed positive box, for example $0.01/m \le p_i \le 2/m$, contains every competitive assignment for all sufficiently large m. This first box is essential before applying (3.8).

Set $P_2 = \sum_i p_i^2$. Conditional Jensen in each centroid direction and weighted Cauchy give the assignment-independent perspective bound

$$A_2(f) \ge \frac{(W - P_2)_+^2}{2P_2}. \tag{3.9}$$

Its positive part makes the expression jointly convex in (W, P_2). The tangent at $(w, 1/m)$, with $\ell = mw - 1$, is

$$A_2(f) \ge F_0 - \frac{\ell}{r}D + c_{m,r}\delta, \quad F_0 = \frac{\ell^2}{2m}, \quad c_{m,r} = \ell\left(\frac{1}{r} - \frac{\ell+2}{2}\right) \ge 0.$$

The exact regular second-chaos identity and $\operatorname{Var}(\max_i Z_i) = O(1/\log m)$ give $0 \le A_2(g) - F_0 \le C/m$. Keeping the quadratic term in the parity comparison therefore gives

$$(1 - r\ell)D + r^2 c_{m,r}\delta \le Cr^2/m + r^3 + r^4.$$

Meanwhile (3.8), once $CrL^2 \le 1/2$, gives $D \ge \delta/2 + r(V(p) - W) \ge 0$. Since $r \le 1/(m-1)$, this proves

$$\delta \le 2D, \qquad 0 \le D \le Cr^2/m, \qquad |W - w| \le D/r. \tag{3.10}$$

It follows that $\max_i |mp_i - 1| \le C\sqrt{m}\, r = o(1)$, and hence the required box $[1/(2m), 3/(2m)]$ holds.

Finally, rank-free offset duality and a physical-graph Cheeger estimate give $\|h_U\|^2 \le CL^2\mathcal{R}(B)$ on the bounded row class. To obtain the deficit dependence, combine the lower fixed-score mass Taylor bound $F_{pp} \succeq -CLI$ with (3.8), then test $p - U = -h_U/(CL^2)$. The halfspace bound $\|h_U\|_\infty \le CL/m$ keeps this test inside the positive mass box. Coincident sites are handled by value/offset limits with equal offsets on duplicate groups. Testing the balanced dual at the actual masses now gives

$$\mathcal{R}(B) \le \Delta/2 + CL\sqrt{\mathcal{R}(B)\delta}, \qquad 0 \le \mathcal{R}(B) \le D/r.$$

The argument uses neither stationarity nor existence of a minimizer of the full-noise functional.

4 Calibrated diagonal derivatives on a coordinate box

This section proves the derivative estimate of [11]. The distinction between a small maximum coordinate and a small Frobenius norm is essential.

Proposition 4.1. *Fix $A > 0$. Let $\sum_i \xi_i = 0$ and $\max_i|\xi_i| \le A$. Set $v_i(t) = 1 + t\xi_i$, $\sigma_i = \sqrt{v_i}$, and choose the unique mean-zero offsets $h(t)$ such that the independent scores $\sigma_i Z_i + h_i$ give every label probability $1/m$. For all sufficiently large m, on $0 \le t \le c_A/L^2$, define*

$$\begin{aligned} F_i(x) &= \Phi((x-h_i)/\sigma_i), & f_i(x) &= \partial_x F_i(x), & a_i(x) &= f_i(x)/F_i(x), \\ W_0(x) &= \prod_i F_i(x), & A_h(x) &= \sum_i a_i(x), & \rho_{ij}(x) &= W_0(x)a_i(x)a_j(x), \\ \omega_{ij} &= \int_{\mathbb{R}} \rho_{ij}(x)\,\mathrm{d}x, & \ell_i(t) &= \sum_j \omega_{ij} = \int_{\mathbb{R}} W_0 A_h a_i\,\mathrm{d}x. \end{aligned}$$

The analytic quantity ω_{ii} is included in ℓ_i but is not a physical facet weight. Put $M_1 = m^{-1}\sum_i|\xi_i|$ and $M_2 = m^{-1}\sum_i \xi_i^2$. Then

$$c\alpha/m \le \omega_{ij} \le C\alpha/m, \tag{4.1}$$

$$|h_i'| \le C\sqrt{L}(|\xi_i| + M_1), \qquad |h_i''| \le CL^{3/2}(\xi_i^2 + M_2), \tag{4.2}$$

$$|\ell_i''(t)| \le C\alpha L^2(\xi_i^2 + M_2). \tag{4.3}$$

The constants depend only on A, not on $\|\xi\|_2$. If T has density proportional to $\phi(x)^2\Phi(x)^{m-2}$, then

$$\ell_i(0) = \alpha, \qquad \ell_i'(0) = -\alpha\eta_m\xi_i, \qquad \eta_m = \frac{m - 1 - m\operatorname{Var} T}{2(m-2)} \in [1/4, 1/2]. \tag{4.4}$$

In particular,

$$|\ell_i(t) - \alpha + \alpha\eta_m t\xi_i| \le \tfrac{1}{2}C\alpha L^2 t^2(\xi_i^2 + M_2). \tag{4.5}$$

4.1 Uniform threshold moments

We first prove the analytic estimate needed for all differentiations.

Lemma 4.2. *For a fixed small $\delta_0 > 0$, suppose*

$$\epsilon := \max_i |v_i - 1| \le \delta_0/L, \qquad H := \max_i |h_i| \le \delta_0/\sqrt{L}. \tag{4.6}$$

Balance is not required here. For every fixed pair of nonnegative integers p, q, uniformly in all i, j, including $i = j$,

$$\omega_{ij} \asymp \alpha/m, \qquad \mathbb{E}_{ij}\left[(m \max_k a_k)^p (1 + |x|)^q\right] \le C_{p,q} L^{(p+q)/2}, \tag{4.7}$$

where $\mathbb{E}_{ij}$ is expectation under ρ_{ij}/ω_{ij}.

Proof. Write

$$Z_0 = \int \phi(x)^2 \Phi(x)^{m-2}\,\mathrm{d}x = \frac{\mu_m}{m(m-1)} = \frac{\alpha}{m} \asymp \frac{\sqrt{L}}{m^2}. \tag{4.8}$$

The identity follows by integration by parts in $\mu_m = m\int x\phi(x)\Phi(x)^{m-1}\,\mathrm{d}x$. The order $\mu_m \asymp \sqrt{L}$ follows from the Gaussian union bound for the upper estimate and the usual Mills lower bound at a fixed fraction of $\sqrt{2\log m}$ for the lower estimate.

For $x \ge 2H$, set $z = (x + H)/\sigma_{\min} \ge 0$. Every standardized argument $y_k = (x - h_k)/\sigma_k$ is nonnegative and belongs to $[z - C\epsilon z - CH, z]$. Comparing Gaussian densities gives

$$\rho_{ij}(x)\,\mathrm{d}x \le C\phi(z)^2\Phi(z)^{m-2} e^{C\epsilon z^2 + CHz + CH^2}\,\mathrm{d}z, \tag{4.9}$$

$$\max_k a_k(x) \le C\frac{\phi(z)}{\Phi(z)} e^{C\epsilon z^2 + CHz + CH^2}. \tag{4.10}$$

For $i = j$ the extra factor F_i^{-1} is at most 2, so the same estimates apply. For a lower weight bound instead use $y = (x - H)/\sigma_{\max}$ on $x \ge 2H$. The lower density is a constant times $\phi(y)^2\Phi(y)^{m-2}e^{-C\epsilon y^2 - CHy - CH^2}$. For $i = j$, use $\Phi(y)^{m-1} \ge \frac{1}{2}\Phi(y)^{m-2}$. The omitted interval near zero has exponentially small baseline mass. The baseline law has $\mathbb{E}T^2 \le CL$: below a fixed multiple of $\sqrt{L}$ this is immediate, and above it the factor ϕ^2, divided by (4.8), gives an integrable Gaussian tail; the negative half-line is exponentially suppressed by Φ^{m-2}. Jensen's inequality therefore bounds the lower integral below by cZ_0.

For the upper moments, put $s = 1 - \Phi(z)$ on $z \ge 0$. Mills bounds give $\phi(z) \le Cs\sqrt{\log(e/s)}$ and $z^2 \le 2\log(1/s)$. Also $CHz \le CH\sqrt{L}/2 + CHz^2/(2\sqrt{L})$. Thus (4.9)–(4.10), including the absolute Jacobian $|\,\mathrm{d}z| = |\,\mathrm{d}s|/\phi(z)$, bound the desired unnormalized moment by

$$C_{p,q} m^p \int_0^{1/2} (1 - s)^{m-p-2} s^{p+1-2\theta} [\log(e/s)]^{(p+q+1)/2}\,\mathrm{d}s, \quad 0 \le \theta \le C_{p,q}\delta_0/L. \tag{4.11}$$

For large m, substitution $u = ms$ and splitting at $u = 1$ give

$$\int_0^{1/2} (1 - s)^{m-p-2} s^{p+1-2\theta} [\log(e/s)]^b\,\mathrm{d}s \le C_{p,q} m^{-p-2+2\theta} L^b.$$

On $u \le 1$ use integrability of a fixed positive power times $\log(1/u)^b$; on $u \ge 1$ use $(1 - s)^{m-p-2} \le e^{-ms/2}$. Since $m^{2\theta}$ is bounded, division by Z_0 proves (4.7) on $x \ge 2H$.

On $x \le -2H$, the product W_0 is at most $2^{-m}e^{-cmx^2}$ and every hazard is at most $C(1 + |x|)$. On $[-2H, 2H]$, every F_i is bounded by a fixed number below one, and the hazards are bounded. These two regions contribute at most $C_{p,q} m^p c_0^m$ for some $c_0 < 1$, negligible relative to Z_0. This also verifies all negative-threshold tails and completes the proof. □

4.2 A componentwise inverse and the calibrated acceleration

The balanced offset exists uniquely in the mean-zero gauge: the Gaussian offset dual is coercive there, its gradient is the mass vector minus U, and its Hessian is the complete Laplacian with positive weights ω_{ij}. These integral weights are smooth for positive variances, so the implicit function theorem gives smooth $h(t)$.

Lemma 4.3. *If $c\alpha/m \le \omega_{ij} \le C\alpha/m$ for $i \neq j$, the associated Laplacian $\mathcal{L}$ satisfies, for mean-zero b,*

$$|(\mathcal{L}^{-1}b)_i| \le \frac{C}{\alpha}\left(|b_i| + \frac{1}{m}\sum_j |b_j|\right). \tag{4.12}$$

Proof. Subtract $c\alpha/m$ from every edge weight. The remainder is a graph Laplacian $\mathcal{L}_0$, and on $\mathbf{1}^\perp$, $e^{-s\mathcal{L}} = e^{-c\alpha s}e^{-s\mathcal{L}_0}$. The second semigroup is doubly stochastic, hence contracts ℓ^1. Integration gives $\|\mathcal{L}^{-1}b\|_1 \le \|b\|_1/(c\alpha)$. For $y = \mathcal{L}^{-1}b$, the row equation $\deg_i y_i = b_i + \sum_{j\neq i}\omega_{ij}y_j$, together with $\deg_i \ge c\alpha$ and the upper weight bound, proves (4.12). □

Define the threshold velocities and logarithmic density derivatives

$$b_i(x) = h_i' + \frac{\xi_i}{2v_i}(x - h_i), \tag{4.13}$$

$$L_i(x) = \partial_t \log f_i = -\frac{\xi_i}{2v_i} + \frac{\xi_i(x-h_i)^2}{2v_i^2} + \frac{h_i'(x-h_i)}{v_i}. \tag{4.14}$$

The density transport equation $\partial_t f_i = -\partial_x(f_i b_i)$ and integration by parts in the winning-probability integral give

$$0 = p_i' = \sum_{j\neq i}\int \rho_{ij}(b_i - b_j)\,\mathrm{d}x. \tag{4.15}$$

While (4.6) holds, the threshold moment lemma and Lemma 4.3 solve this equation for h' and give the first estimate in (4.2).

This estimate closes the box by continuation. Starting with $h(0) = 0$, it gives $\|h(t)\|_\infty \le C_A t\sqrt{L}$. Choose a bootstrap bound twice this size and then choose c_A small. For $t \le c_A/L^2$ the offset and variance bounds lie strictly inside (4.6). The constants in that box do not depend on the bootstrap constant. Thus the first derivative estimate and the weight comparison hold on the whole interval, without smallness of $\|h\|_2$ or $\|\xi\|_2$.

For the second derivative set

$$S(x) = \sum_k a_k b_k, \qquad P_{ij} = L_i + L_j + a_i b_i + a_j b_j - S, \tag{4.16}$$

$$c_i(x) = -\frac{\xi_i^2(x-h_i)}{2v_i^2} - \frac{\xi_i h_i'}{2v_i}. \tag{4.17}$$

Then $\partial_t \log \rho_{ij} = P_{ij}$, also for $i = j$, and $b_i' = h_i'' + c_i$. For every fixed finite k, the moment lemma gives

$$\|b_i\|_{L^k(ij)} \le C_k\sqrt{L}(|\xi_i| + M_1), \tag{4.18}$$

$$\|P_{ij}\|_{L^k(ij)} \le C_k L(|\xi_i| + |\xi_j| + M_1). \tag{4.19}$$

For the summed term use $\sum_k a_k(|\xi_k| + M_1) \le 2m\max_k a_k M_1$. This step retains the contributions of all other labels without losing a factor m.

Differentiating (4.15) yields exactly

$$\mathcal{L}h'' = -b^{(2)}, \qquad b_i^{(2)} = \sum_{j\neq i} \int \rho_{ij}\{P_{ij}(b_i - b_j) + c_i - c_j\}\,\mathrm{d}x. \tag{4.20}$$

Hölder's inequality, (4.19), and $M_1^2 \le M_2$ give $|b_i^{(2)}| \le C\alpha L^{3/2}(\xi_i^2 + M_2)$ and $m^{-1}\sum_i |b_i^{(2)}| \le C\alpha L^{3/2} M_2$. The right side has mean zero by pairwise antisymmetry. Applying Lemma 4.3 proves the asserted bound for h_i''.

4.3 Second derivatives of the hazard coefficients

Since $\partial_t^2 \rho_{ij} = \rho_{ij}(P_{ij}^2 + P_{ij}')$, it remains to estimate the two terms. Differentiating (4.14) gives

$$L_i' = \frac{\xi_i^2}{2v_i^2} - \frac{\xi_i^2(x-h_i)^2}{v_i^3} - \frac{2\xi_i h_i'(x-h_i)}{v_i^2} + \frac{h_i''(x-h_i)}{v_i} - \frac{(h_i')^2}{v_i}.$$

Its conditional L^k norm is at most $C_k L^2(\xi_i^2 + M_2)$. Moreover

$$a_i' = a_i(L_i + a_i b_i), \qquad (a_i b_i)' = a_i\{(L_i + a_i b_i)b_i + b_i'\}.$$

The same bound holds for this last expression. For S' first sum pointwise using

$$\sum_k a_k \xi_k^2 \le m \max_k a_k M_2, \qquad \sum_k a_k (|\xi_k| + M_1)^2 \le Cm \max_k a_k M_2.$$

An additional factor $\max a_k$ handles the terms with a_k^2. Lemma 4.2 then yields $\|S'\|_{L^k(ij)} \le C_k L^2 M_2$. Consequently

$$\mathbb{E}_{ij}(P_{ij}^2 + |P_{ij}'|) \le CL^2(\xi_i^2 + \xi_j^2 + M_2). \tag{4.21}$$

All derivatives are dominated by the integrable functions just estimated, so differentiation under the integral is legitimate. Summing (4.21) with $\omega_{ij} \le C\alpha/m$, including $j = i$, proves (4.3).

4.4 The exact regular coefficient

At $t = 0$, all variances are one and all offsets are zero. Equation (4.15) gives $h_i'(0) = -(\mathbb{E}T)\xi_i/2$. Put $M = \mathbb{E}T$ and $a(x) = \phi(x)/\Phi(x)$. Then

$$\partial_t \log a_i|_{t=0} = \frac{(x + a(x))(x - M) - 1}{2}\,\xi_i =: k(x)\xi_i.$$

Because $\sum_i \xi_i = 0$, both $\partial_t W_0$ and $\partial_t A_h$ vanish at zero. Thus $\ell_i'(0) = \alpha \mathbb{E}_T k(T)\xi_i$, and $\ell_i(0) = mZ_0 = \alpha$.

The logarithmic derivative of the tie density is $-2x + (m-2)a(x)$. Integration by parts against $x - M$ gives

$$-2\operatorname{Var} T + (m-2)\operatorname{Cov}(T, a(T)) = -1.$$

It follows that $\mathbb{E}k(T) = [m \operatorname{Var} T - m + 1]/[2(m-2)]$. The tie density has logarithmic second derivative at most -2, so $\operatorname{Var} T \le 1/2$. For the reverse bound, decompose the independent regular scores into their common mean $\overline{Z}$ and their centered vector. Conditioning on $Z_i = Z_j$ and on their being maximal depends only on the centered vector. The raw threshold is $T = \overline{Z} + T_H$, with independent $\operatorname{Var}\overline{Z} = 1/m$. Hence $\operatorname{Var} T \ge 1/m$. This proves (4.4) and its coefficient interval. Integrating (4.3) twice proves (4.5), completing Proposition 4.1.

Remark 4.4 (Normalization check). The physical degree is $\ell_i - \omega_{ii}$. At zero, $\omega'_{ii} = -2\alpha\eta_m\xi_i/m$. Therefore Price's formula gives, in variance coordinates,

$$F''_v[\xi,\xi] = -\frac{\alpha}{4}\left(\frac{m-1}{m} - \operatorname{Var} T\right)\|\xi\|_2^2.$$

In standard-deviation coordinates the corresponding balanced Hessian is $\alpha \operatorname{Var} T\|\xi\|_2^2$. These local checks do not replace the uniform coordinatewise remainder proved above.

5 Global summed radial coercivity

We now prove the supporting-covariance theorem of [12]. Only the supporting point is near the regular covariance; the target may be singular and far from it.

Theorem 5.1. *Fix $A \geq 1$. For sufficiently large m, every centered score matrix B with $\|B\|_{\mathrm{F}}^2 = w$ and $|B_i|^2 \leq A\alpha^2$ satisfies*

$$\mathcal{R}(B) \geq \frac{c_A\alpha^2}{L^2}\sum_i\left(\frac{|B_i|^2}{\alpha^2} - \frac{n}{m}\right)^2, \tag{5.1}$$

$$\sum_i(|B_i|^2 - w/m)^2 \leq C_A\frac{L^3}{m^2}\mathcal{R}(B). \tag{5.2}$$

No lower row bound, pair separation, rank, or proximity assumption is required.

Proof. For positive v_i let $D = P\operatorname{diag}(v_i)P$. Independent scores $\sqrt{v_i}Z_i$, after subtracting their common sample mean, realize D. This subtraction changes neither the allocation nor its expected value. Using balanced offsets and the notation of Proposition 4.1, the mass-offset Laplacian is

$$\mathcal{L}_D = P\operatorname{diag}(\ell_i)P - H_D, \qquad H_D = \int W_0(Pa)(Pa)^T\,\mathrm{d}x \succeq 0. \tag{5.3}$$

Indeed $\operatorname{diag}(\ell) - \int W_0 aa^T$ is exactly the ordinary Laplacian, which already annihilates $\mathbf{1}$.

At this positive supporting point, Analytic interface 2.1 gives $\nabla\Gamma(D) = \mathcal{L}_D/2$. Homogeneity of degree $1/2$ gives $\Gamma(D) = \operatorname{tr}(\mathcal{L}_D D)$. Concavity therefore implies, for every centered PSD target C,

$$\begin{aligned}\Gamma(C) &\leq \Gamma(D) + \tfrac{1}{2}\operatorname{tr}[\mathcal{L}_D(C-D)] = \tfrac{1}{2}\operatorname{tr}[\mathcal{L}_D(C+D)]\\ &\leq \tfrac{1}{2}\sum_i \ell_i(C_{ii}+D_{ii}).\end{aligned} \tag{5.4}$$

The discarded term is $-\operatorname{tr}[H_D(C+D)]/2 \leq 0$. This is an exact PSD sign, not an entrywise comparison. Continuity gives the same support inequality for singular C.

Set $C = BB^T/\alpha^2$, so $\operatorname{tr} C = n$ and $C_{ii} \leq A$. Write

$$z_i = C_{ii} - n/m, \qquad \kappa = 1 - 2/m, \qquad \xi_i = z_i/\kappa.$$

Then $\sum_i \xi_i = 0$, $\|\xi\|_\infty \leq A_1 := 3(A+1)$, and

$$D_t = P + tP\operatorname{diag}(\xi)P, \qquad \operatorname{tr} D_t = n, \qquad (D_t)_{ii} = n/m + \kappa t\xi_i = n/m + tz_i. \tag{5.5}$$

Let c_H, C_H be the constants from Proposition 4.1 for A_1. Choose

$$0 < b_A \leq \min\left\{c_H, 1, \frac{1}{2A_1}, \frac{1}{24C_H(A+2)}\right\}, \qquad t = b_A/L^2.$$

The supporting covariance is positive definite on $\mathbf{1}^\perp$, its diagonal is at most 2, and the whole segment lies in the proved coordinate box. Taylor's formula gives

$$\ell_i(t) = \alpha - \alpha\eta_m t\xi_i + e_i, \qquad |e_i| \le \tfrac{1}{2}C_H\alpha L^2t^2(\xi_i^2 + m^{-1}\|\xi\|^2).$$

Inserting this in (5.4), the constant term is $\alpha n = \mu_m$. The linear term pairs all diagonal errors at once:

$$\sum_i \xi_i(C_{ii} + (D_t)_{ii}) = \kappa(1+t)\|\xi\|^2.$$

The absolute remainder is at most $C_H(A+2)\alpha L^2t^2\|\xi\|^2/2$. Since $\eta_m \ge 1/4$ and $\kappa \ge 1/3$,

$$\begin{aligned}\mu_m - \Gamma(C) &\ge \alpha\left\{\frac{\eta_m\kappa t(1+t)}{2} - \frac{C_H(A+2)L^2t^2}{2}\right\}\|\xi\|^2 \\ &\ge \frac{\alpha\kappa t}{16}\|\xi\|^2 \ge \frac{\alpha b_A}{16L^2}\|z\|^2.\end{aligned}$$

For the middle inequality, after factoring αt, the error is at most $1/48$, and $\kappa/8 - 1/48 \ge \kappa/16$. Finally $\mathcal{R}(B) = w - \alpha\Gamma(C) = \alpha[\mu_m - \Gamma(C)]$. This proves (5.1); multiplying by α^4 and using $\alpha^2 \asymp L/m^2$ gives (5.2). □

Corollary 5.2 (Bounded unnormalized scales)**.** *Fix $0 < s_0 \le s_1 < \infty$ and A. If B is centered, $s = \|B\|/\sqrt{w} \in [s_0, s_1]$, and $|(B/s)_i|^2 \le A\alpha^2$, then*

$$\sum_i (|B_i|^2 - W/m)^2 + \sum_i (|B_i|^2 - w/m)^2 \le C\frac{L^3}{m^2}\mathcal{R}(B), \tag{5.6}$$

$$E_{\mathrm{rad}} := \sum_i (|B_i| - \beta)^2 \le CL^2\mathcal{R}(B). \tag{5.7}$$

Proof. Put $\widehat{B} = B/s$. Equation (2.3) is exact. The vector $(|B_i|^2 - W/m)_i$ equals s^2 times its hatted version, so Theorem 5.1 bounds its squared norm by $Cs^3\alpha^2L^2\mathcal{R}(B)$. Orthogonal decomposition into the mean and mean-zero parts gives

$$\sum_i (|B_i|^2 - w/m)^2 = \sum_i (|B_i|^2 - W/m)^2 + \frac{(W-w)^2}{m}.$$

The last term is $w^2(s+1)^2(s-1)^2/m \le 2w(s+1)^2\mathcal{R}(B)/m$. This proves (5.6). Finally $(|B_i| - \beta)^2 \le (|B_i|^2 - \beta^2)^2/\beta^2$ and $\beta^2 \asymp L/m^2$ give (5.7). □

6 An exactly centered equal-row test matrix

The matrix constructed next is only a test object. We will neither route the assignment by it nor claim it is a matrix of actual centroids.

Lemma 6.1. *Suppose $\widehat{B}$ is centered, $\|\widehat{B}\|^2 = w$, $\|\widehat{B}\|_{\mathrm{op}}^2 \le C/m$, and $\max_i|\widehat{B}_i|/\beta - 1| = o(1)$. Let $E = \sum_i(|\widehat{B}_i| - \beta)^2$ and suppose $E \le CrL^2/m$ for $0 < r \le 1/(m-1)$. For sufficiently large m there is a matrix Z, in the same row span, such that*

$$\sum_i Z_i = 0, \qquad |Z_i| = \beta, \qquad \|Z - \widehat{B}\|^2 \le CE. \tag{6.1}$$

Proof. Write $u_i = \widehat{B}_i/|\widehat{B}_i|$. The row lower bound and operator estimate give $\|m^{-1}\sum_i u_i u_i^T\|_{\mathrm{op}} \le C/L$. On the row span consider $\Psi(c) = m^{-1}\sum_i|\widehat{B}_i - c|$. For $|c| \le c_0\beta$, distances are comparable to β, and

$$\left\| \frac{1}{m}\sum_i \frac{(\widehat{B}_i - c)(\widehat{B}_i - c)^T}{|\widehat{B}_i - c|^2} \right\|_{\mathrm{op}} \le C/L + C|c|^2/\beta^2.$$

Choose a fixed sufficiently small c_0. Then $\nabla^2\Psi \succeq c\beta^{-1}I$ on that ball. At zero, centering gives

$$g_0 := \|\nabla\Psi(0)\| \le \frac{C\|\widehat{B}\|_{\mathrm{op}}\sqrt{E}}{m\beta^2} \le C\frac{\sqrt{m}\sqrt{E}}{L} \le C\sqrt{r} = o(1).$$

The radial derivative is positive on the ball boundary for large m; therefore Ψ has an interior minimizer c, with $|c| \le C\beta g_0$ and $\sum_i(\widehat{B}_i - c)/|\widehat{B}_i - c| = 0$. Define $Z_i = \beta(\widehat{B}_i - c)/|\widehat{B}_i - c|$. This gives the first two assertions in (6.1). Moreover

$$m|c|^2 \le CE/L.$$

Compare Z_i first with βu_i. Radial projection is uniformly Lipschitz on these points away from zero, so the squared comparison cost is $Cm|c|^2$; the squared cost from βu_i to $\widehat{B}_i$ is exactly E. This proves the final assertion. All operations take place in the row span. □

Remark 6.2. The proof of Lemma 6.1 is the geometric-median construction from [6], with its small-gradient condition supplied here by the radial bound. Neither duplicate rows of Z nor rank loss will cause a problem in the quadratic test below.

7 Finite covariance gaps and arbitrary PSD endpoints

We supply the quantitative ray argument used in both noise branches. The full reconstruction, including covariance tilts, is [30]. Write $b = \Phi^{-1}(1 - 1/m)$, so $b^2 \asymp L$ for large m. At P, every physical score-facet weight is $w_0 = \alpha/m$.

7.1 Facet bootstrap near the regular covariance

The regular facet Gaussian has mean of size $O(b)$ and covariance S_e satisfying $\|S_e - P\|_{\mathrm{F}} \le C$. The latter bound has no ambient-dimension factor: the normal variance is zero, one distinguished variance lies in $[0, 1]$, and the remaining $m-3$ variances are $1-O(b^2/m)$. These formulas follow from the tied-maximum density proportional to $\phi(t)^2\Phi(t)^{m-2}$ and the truncated independent coordinates conditional on the tied value.

For $K = P + E$ with $\epsilon = \|E\|_{\mathrm{F}} \le 1/8$, a shifted facet density relative to its regular counterpart is $\exp(c + q(y))$, where

$$q(y) = \tfrac{1}{2}y^T(P - K^{-1})y + (K^{-1}h)\cdot y, \quad c = -\tfrac{1}{2}\log\det_H K - \tfrac{1}{2}h^T K^{-1}h.$$

The trace cancellation $|\operatorname{tr}_H(P - K^{-1}) - \log\det_H K| \le C\epsilon^2$ removes any $\sqrt{m}\epsilon$ contribution. The tilted tangent potential remains uniformly convex. Brascamp–Lieb and differentiation of the mean give

$$\mathrm{Var}_t(q) \le C(\epsilon^2 b^2 + |h|^2), \qquad |\mu_1 - \mu_0| \le C(\epsilon b + |h|).$$

Thus the facet weight ratio lies between e^{-A} and e^A, where $A = C(\epsilon b^2 + b|h| + |h|^2 + \epsilon^2 b^2)$.

Along $K_t = P + tE$, the balancing offset solves a complete-graph Laplacian equation. While weights lie between $w_0/2$ and $2w_0$, the centered graph gap is at least $mw_0/2$ and its forcing energy is at most $Cmw_0b^2\epsilon^2$. Hence $|h_t'| \le Cb\epsilon$. Choose the bootstrap coefficient first, then an absolute $c > 0$ such that $\epsilon \le c/b^2$ closes the weight and offset bounds strictly. This proves uniform facet control on $\|K - P\|_{\mathrm{F}} \le c/L$.

7.2 Normal strain and propagation along a ray

For a centered symmetric E, the exact finite identity is

$$\sum_{i<j}[(e_i - e_j)^T E(e_i - e_j)]^2 = m\sum_i E_{ii}^2 + (\operatorname{tr} E)^2 + 2\|E\|_{\mathrm{F}}^2. \tag{7.1}$$

Apply (3.2) on the preceding ball. Edge denominators are bounded above by three, so $-D^2\Gamma(K)[E,E] \ge w_0\|E\|_{\mathrm{F}}^2/12$. The derivative at P is $D\Gamma(P)[E] = (\alpha/2)\operatorname{tr} E$.

Now let $K = BB^T/\alpha^2$ be any centered PSD covariance and put $e = \|K - P\|_{\mathrm{F}}$. Invariance of Gaussian realizations gives

$$\mathcal{R}(B) = \alpha\{\Gamma(P) + D\Gamma(P)[K - P] - \Gamma(K)\}.$$

If $e > 0$, integrate curvature only on $0 \le t \le s = \min\{1/2, c/(Le)\}$ along $P + t(K - P)$. The rest of the ray has a nonnegative curvature measure by concavity. Taylor's integral weight is $1 - t$, at least $1/2$ on this initial interval. Consequently

$$\mathcal{R}(B) \ge \frac{c\alpha^2 e^2}{m}\min\{1, c/(Le)\}. \tag{7.2}$$

For a singular endpoint, integrate first to $T < 1$ and then let $T \uparrow 1$ by value continuity. No global lower bound on the competitor's eigenvalues is used.

Procrustes gives $q = d/\alpha$ with $q^2 = \sum_{i=1}^{m-1}(\sqrt{\lambda_i(K|_H)} - 1)^2$, with eigenvalues taken on $H = \mathbf{1}^\perp$ and including any zeros on that subspace. Since $e \ge q$ and $x^2\min\{1, c/(Lx)\}$ is increasing, (7.2) also holds with e replaced by q. Competitive norm localization gives $q \le C\sqrt{m}$; in the exterior $d > 1/m$, one has $q \ge c/\sqrt{L}$. Hence

$$\mathcal{R}(B) \ge cq/m^3 \ge c/(m^3\sqrt{L}), \qquad d^2 \le Cm^{3/2}L\mathcal{R}(B). \tag{7.3}$$

For the local score ball $d \le A/m$, a correspondingly short initial segment gives $d^2 \le C_A m\sqrt{L}\mathcal{R}(B)$. Retaining the diagonal part in (7.1) also gives the stronger normal energy used in the local unequal-row transfer.

7.3 A single row cannot collapse

At fixed trace, symmetrize a covariance over permutations fixing one label. It becomes $K_a = auu^T + b(P - uu^T)$ with $b = (m - 1 - a)/(m - 2)$, where $a = |\widehat{B}_i|^2/\beta^2$. For edges incident to that label, the squared edge length is $a + 1$ and its derivative is one. Normal strain and Gaussian isoperimetry of that balanced cell therefore imply

$$-4\frac{\mathrm{d}^2}{\mathrm{d}a^2}\Gamma(K_a) \ge \frac{I(1/m)}{(a+1)^{3/2}}, \qquad I(p) = \phi(\Phi^{-1}(p)).$$

The tangent at $a = 1$ vanishes. Integration gives

$$\mathcal{R}(\widehat{B}) \ge \frac{\alpha I(1/m)}{2\sqrt{2}}(\sqrt{1+a} - \sqrt{2})^2.$$

Continuity covers $a = 0$ and $a = m - 1$. On any fixed row-cap class this proves (2.7). Together with competitive localization it supplies precisely the row lower bounds required by the equal-row proxy.

8 The local comparison at every perturbation scale

We reconstruct the local interface (2.9). The complete affine trace, moving-boundary, and cap estimates are supplied in [31, 33, 34, 35]; they apply to the actual centroids of arbitrary fractional assignments. Fix A and suppose $d = d(B) \le A/m$ and $p_i \in [1/(2m), 3/(2m)]$. Write $z = p - U$ and $R = \mathcal{R}(B)$. After alignment with a regular matrix, all $m - 1$ singular values of B on $H = \mathbf{1}^\perp$ lie in $[\alpha/2, 2\alpha]$ for large m, since $\alpha \asymp \sqrt{L}/m$. Full score rank is therefore a consequence of this local regime, rather than a global assumption on competitors.

8.1 Conditioned affine fans and the signed affine comparison

Let f^0 be the affine winner with scores B and offsets giving the actual masses p, and let M be its centroid matrix. Gaussian integration by parts gives $M = \mathcal{L}B$. The centroid entropy bound, the pseudoinverse of B, and isoperimetry on unions of cells give bounded graph degrees and a gap $\lambda_2(\mathcal{L}) \ge c/L$.

Every specified pair facet has intrinsic Gaussian probability at least m^{-C}, with no dimension-dependent small-ball argument. To see the key construction, normalize scores and offsets by α. Their singular values lie in $[1/2, 2]$, and pair halfspaces give $H = \max_i |h_i| \le C\sqrt{L}$. For labels i, j, choose $T = H + 32\sqrt{L}$, set $q_i = T - h_i$, $q_j = T - h_j$, and set every other $q_k = -(q_i + q_j)/(m - 2)$. Then $q \in H$, $\|q\|^2 = O(L)$, and its minimum-norm preimage has squared norm $O(L)$. At that point the two selected utilities dominate the others. A tangential Gaussian union bound followed by the Cameron–Martin inequality gives the polynomial facet probability. Only two score coordinates are made large.

This facet bound and convex Gaussian conditioning give conditional moments of fixed-degree polynomials with powers of L, independent of N. The normal Mehler row on a hyperplane at coordinate a satisfies

$$K_t(a)^2 = (1 - t^4)^{-1/2} \exp\{t^2 a^2/(1 + t^2)\}.$$

At $t^2 = 1/(2q - 1)$, its product with $\phi(a)^{1/q}$ has no exponential a dependence. Combined with tangential hypercontractivity at $q = CL$, this gives the dimension-free affine trace estimate in [33]. The argument separates at most two normal directions; it never estimates a full-N pointwise Mehler determinant.

To compare second chaos along a balanced affine path, the adjoint Laplacian equation removes the entire offset-acceleration coefficient. A coherent affine velocity moves all interfaces and their intersections. Its surface divergence and the derivative of inverse edge length combine into Gaussian divergence; junction contributions cancel under this same velocity. At the regular reference, the pair potential difference factors through $X_i - X_j$ and vanishes on the regular facet. On a perturbed facet the factor is the negative score displacement divided by α, providing the deficit factor in the remaining transport terms.

The unequal-row transfer needs more than an unweighted score Hessian bound. First normalize the scores to squared Frobenius norm w, and write their covariance divided by α^2 as $P + E$. Thus $\operatorname{tr} E = 0$. Decompose $E = E_d + E_s$, where E_s has zero diagonal and $E_d = mP \operatorname{diag}(\operatorname{diag} E)P/(m-2)$.

If $a = \|E_d\|$ and $b = \|E_s\|$, the retained diagonal normal energy gives

$$R \geq \frac{c_A \alpha^2}{m\sqrt{L}}(ma^2 + b^2).$$

The rectangle error is bounded by $C_A \alpha^2 L^{10} a(a+b)$. The inequalities $a^2 \leq (ma^2+b^2)/m$ and $ab \leq (ma^2+b^2)/(2\sqrt{m})$ pay this error relative to R. The exact radial identity (2.3) restores the original score scale and retains its nonnegative radial square. The equal-row input uses only the credited common-threshold comparison of Analytic interface 2.7; its affine offset terms are canceled before applying that comparison. No stochastic domination for unequal thresholds is asserted.

For higher chaos, factor the regular tail potential through a fixed OU smoothing time and use the odd pair difference on the reference facet. The same adjoint and coherent second variation yield a bound $C_A r^3 L^{10} \|Z\|^2$ on its score Hessian. Finite Hermite cutoffs obey bounds uniform in the cutoff; the residual geometric multipliers are below one, so the potentials and boundary derivatives converge. The resulting affine comparison is

$$\begin{aligned} S_{-r}(f^0) - S_{-r}(g) \geq [1 - C_A(r^2 m L^7 + r^3 L^{10})]\|z\|^2 + r(w - \|M\|^2) \\ - C_A(r^2\sqrt{m} L^{13} + r^3 m L^{11}) R. \end{aligned} \tag{8.1}$$

The derivations of the boundary Hessian and of the infinite signed-tail passage are substantive analytic parts of [33, 35]; (8.1) is not inferred from a numerical truncation.

8.2 The exact actual-to-affine identity

Put $v = f - f^0$ and $L_{\mathrm{route}} = F_p(B) - \|B\|^2 = \langle B, M - B\rangle \geq 0$. Feasibility of the actual assignment in its mass class proves this inequality. Let $u = T_{\sqrt{r}} \Pi_{\mathrm{odd}\geq 3} v$, and set $q_B = T_{-r}\Pi_{\geq 2} f^0$. Let q_0 denote the corresponding regular potential. Hermite orthogonality, with only nonnegative even squares dropped, gives

$$\begin{aligned} S_{-r}(f) - S_{-r}(f^0) \geq r\|M - B\|^2 + \|u\|_2^2 + 2\{r L_{\mathrm{route}} + \langle v, q_0 + a\rangle\}, \\ a = q_B - q_0 - T_{\sqrt{r}} u. \end{aligned} \tag{8.2}$$

In particular the positive centroid square is retained.

The two-label regular comparison has a positive divided coefficient k_{ij}. With $\eta = O(r^2 L)$ and $\kappa(x) = \exp(-1 - \eta|P_0 x|^2)$, one has $k_{ij} \geq \kappa$. A smooth log-sum-exp range majorant controls $|1 - k_{ij}|$ by e_r. On the old affine cell i, this bounds the last bracket in (8.2) below by minus r times the expected cap

$$\max\{0, \max_{j \neq i}(\psi_i + \psi_j - c_{ij})\}, \qquad \psi_i = \kappa^{-1}\{e_r|\delta l_i| + |a_i|/r\}.$$

The diagonal $j = i$ is excluded: it would create an artificial reward with zero switching cost.

8.3 A quadratic cap with no amplitude floor

Write $D_0 = d + \|z\|$. The trace and restriction estimates imply

$$\|\psi\|_{L^q(\ell^2)} \leq C(r L^{5/2} D_0 + \sqrt{r} L^{3/2}\|u\|_2), \qquad q = C_1 L,$$

and a corresponding weighted facet-square bound $C(r^2 L^5 D_0^2 + r L^3 \|u\|_2^2)$. Radial exponential moments remain bounded: on a convex facet restriction, for a rank-at-most-m projection,

$$\mathbb{E} e^{b|PY|^2} \leq (1 - 2b)^{-m/2} \exp\{b|P\mathbb{E}Y|^2/(1 - 2b)\}.$$

For $b = q\eta$, one has $bm = O(L^2/m)$ and $|\mathbb{E}Y|^2 = O(L)$. There is no dependence on the unused ambient dimensions.

Choose $\theta = C_2 L^4(rD_0 + \sqrt{r}\|u\|_2)$. If $\theta = 0$, the forcing vanishes. Otherwise $v_i = \psi_i/\theta$ has controlled vector and weighted facet moments, and $\theta \le m^{-5/4}$ for large m. For moderate amplitudes $\theta \ge m^{-B}$, the separate source-exit and target-entry margin estimates give the cap bound

$$C\left[m^{1/(q+1)}\theta^{2q/(q+1)} + \frac{c}{m}(C\theta m/c)^q\right] = O_B(\theta^2).$$

Here $q = C_1 L$ is fixed independently of θ. This estimate alone does not give a quadratic bound as $\theta \downarrow 0$ at fixed m.

For smaller amplitudes, the structured forcing has entire-plane L^{32} bounds, including its first normal derivatives, bounded by a fixed power of m after division by θ. The smoothed input norms divided by $r\theta$ remain bounded, so no inverse small-r factor is hidden. Set $M_{\mathrm{cut}} = \theta^{-1/8}$. At good projected points the normal reward width is $O(\theta M_{\mathrm{cut}}/\alpha)$. Taylor and density errors are $O(m^C\theta^3 M_{\mathrm{cut}}^3)$. Points outside the actual facet that enter an old cell lie in a third-label strip of this width; the local singular-value bound controls its tangent altitude. Bad projected points cost $O(m^C\theta M_{\mathrm{cut}}^{-31})$, and normal distances greater than one cost $O(m^C\theta^{32})$. The main facet term is $C\theta^2$. Thus the cap is at most

$$C\theta^2 + Cm^K\theta^{21/8}.$$

Fix $B > 8K/5$, enlarged for the fixed constants. The second term is absorbed for $\theta \le m^{-B}$. Joining the two ranges proves the quadratic estimate at every amplitude. The geometric strip and normal supremum details are supplied in [34].

Substituting in (8.2), and retaining the nonnegative residual coefficient of $\|u\|_2^2$, gives

$$S_{-r}(f) - S_{-r}(f^0) \ge r\|M - B\|^2 - Cr^3L^8(d^2 + \|z\|^2). \tag{8.3}$$

8.4 Cancellation and the local conclusion

The essential algebra is exact:

$$r(w - \|M\|^2) + r\|M - B\|^2 = 2r\{(w + \|B\|^2)/2 - F_p(B)\}. \tag{8.4}$$

Balanced offsets are an admissible dual test at p, so the braces are at least $R + h_U \cdot z$. The deficit-relative offset certificate and Young's inequality give $2h_U \cdot z \ge -R/2 - CL^2\|z\|^2$. Use $d^2 \le C_A m\sqrt{L}R$ to pay the geometric error. Adding (8.1) and (8.3) yields

$$\begin{aligned} S_{-r}(f) - S_{-r}(g) \ge\ & [1 - C_A(rL^2 + r^2mL^7 + r^3L^{10})]\|z\|^2 \\ & + [3/2 - C_A(r\sqrt{m}L^{13} + r^2mL^{11})]rR. \end{aligned} \tag{8.5}$$

Every loss tends to zero uniformly for $0 < r \le 1/(m-1)$. Choose m sufficiently large, depending only on fixed A, to make the two brackets at least $1/2$ and one. This proves (2.9). In the final assembly $A = 1$ is fixed once; neither the threshold nor a constant depends on the assignment or N.

9 The exterior comparison and the full signed objective

We prove the exterior calculation of [13], with its radial premise supplied by Theorem 5.1, using the direct competitive argument [16]. Let f be any measurable assignment satisfying $S_{-r}(f) \le S_{-r}(g)$,

and put $\rho = -r$. Use the notation of Analytic interface 2.3, and write $R = \mathcal{R}(B)$ and $d = d(B)$. If $d \le 1/m$, (2.9) proves the desired conclusion. Throughout this section assume

$$d > 1/m. \tag{9.1}$$

9.1 The normalized cost and its exterior floor

From $0 \le \delta \le 2D$, one has $r|\Delta| = |D - \delta| \le D$; therefore $W/w = 1 + O(r/L)$, including when $W > w$. Since f is admissible for $F_p(B)$, $F_p(B) \ge W$. The balanced optimal offsets are a dual test at p, so

$$R \le \frac{w - W}{2} - h_U \cdot (p - U) \le \frac{\Delta}{2} + CL\sqrt{R\delta}.$$

Here (2.2) is scaled using (2.3), and the entropy bound gives the required fixed row cap. Young's inequality implies, for large m,

$$0 \le R \le \Delta + CL^2\delta \le \Delta + \delta/r = D/r =: J. \tag{9.2}$$

The division by r is indispensable.

Let $q = d/\alpha$ and $e = \|BB^T/\alpha^2 - P\|_{\mathrm{F}}$. If s_i are the singular values of B/α on $\mathbf{1}^\perp$, including zeros, Procrustes minimization gives $q^2 = \sum_i (s_i - 1)^2$ and $e^2 = \sum_i (s_i^2 - 1)^2$; hence $e \ge q$. Also $c/\sqrt{L} \le q \le C\sqrt{m}$ by (9.1) and $W \asymp w$. Equation (2.8) gives

$$J \ge R \ge \frac{cq}{m^3} \ge \frac{c}{m^3\sqrt{L}}, \qquad d^2 \le Cm^{3/2}LR. \tag{9.3}$$

These estimates concern the score remainder, not a replacement of the actual assignment by an affine one.

Put $s = \sqrt{W/w}$ and $\widehat{B} = B/s$. By Corollary 5.2, $E_{\mathrm{rad}}(\widehat{B}) \le CL^2R \le CrL^2/m \le CL^2/m^2$. The radial identity gives $\mathcal{R}(\widehat{B}) \le CR$, and (2.7) yields

$$\max_i \left| \frac{|\widehat{B}_i|^2}{\beta^2} - 1 \right| \le C\sqrt{\frac{m^2R}{L}} \le C\sqrt{\frac{mr}{L}} = o(1).$$

The universal operator bound (2.5) scales to $\|\widehat{B}\|_{\mathrm{op}}^2 \le C/m$. Thus every hypothesis of Lemma 6.1 is proved for this arbitrary competitive assignment. It gives an exactly centered equal-row matrix Z, with median gradient at most $C\sqrt{r}$. Since $\|B - \widehat{B}\|^2 = w(s-1)^2 \le 2R$,

$$E_Z := \|Z - B\|^2 \le CL^2R \le CL^2J. \tag{9.4}$$

9.2 A maximum comparison with an explicit absolute tail

Write $Y_i = Z_i \cdot X/\beta$, $M = \max_i Y_i$, and let M_* be the regular unit-row maximum. Centering gives $M \ge 0$ pointwise. Set $a_0 = (m-1)/m$ and let $\nu = \sqrt{a_0}M + G/\sqrt{m}$, with G independent standard Gaussian. The underlying Gaussian score covariance is $a_0C + \mathbf{1}\mathbf{1}^T/m$, where $C = ZZ^T/\beta^2$. It has unit diagonal and dominates $\mathbf{1}\mathbf{1}^T/m$. The regular version has covariance I_m. Analytic interface 2.7 therefore gives

$$H(t) := \mathbb{P}\{\nu \le t\} - \mathbb{P}\{\max_i Z_i^{\mathrm{iid}} \le t\} \ge 0.$$

Set $\varepsilon_{\mathrm{free}} = \mathbb{E}M_* - \mathbb{E}M \ge 0$. Integration of the CDFs and cancellation of the independent common Gaussian yield

$$\int H = \sqrt{a_0}\varepsilon_{\mathrm{free}}, \qquad \mathbb{E}M_*^2 - \mathbb{E}M^2 = \frac{2}{a_0}\int tH(t)\,\mathrm{d}t.$$

For $t \geq T$, $H(t) \leq m\overline{\Phi}(t)$; for $t \leq -T$, $H(t) \leq \Phi(t)$. Splitting the integral and using $H \geq 0$ gives, for $T \geq 1$,

$$(\mathbb{E}M_*^2 - \mathbb{E}M^2)_+ \leq CT\varepsilon_{\text{free}} + Cme^{-T^2/2}. \tag{9.5}$$

No deficit-relative tail estimate is assumed. With $V = \beta M = \max_i Z_i \cdot X$ and $V_* = \beta M_*$, $\mathbb{E}V_* = w$ and (9.5) becomes

$$(\mathbb{E}V_*^2 - \mathbb{E}V^2)_+ \leq C\beta T(w - \mathbb{E}V) + C\beta^2 me^{-T^2/2}. \tag{9.6}$$

In particular $\mathbb{E}V \leq w$.

9.3 The actual assignment pays its test-score error

Write $z_i(X) = Z_i \cdot X$ and $c_f(X) = V - \sum_i f_i z_i \geq 0$. Both $\mathbb{E}c_f$ and $w - \mathbb{E}V$ are nonnegative, and their sum is exactly

$$w - \langle Z, B\rangle = \frac{\Delta + E_Z}{2} \leq CL^2 J. \tag{9.7}$$

Here and below $\langle Z, B\rangle = \sum_i Z_i \cdot B_i$. This identity is valid even if $\Delta < 0$ and the masses are unequal. Pointwise, $z_i \leq V$ and $V \geq 0$, so

$$V^2 - \sum_i f_i z_i^2 = \sum_i f_i (V - z_i)(V + z_i) \leq 2Vc_f.$$

If $A_X = \max_i |Y_i|$, then $c_f \leq 2\beta A_X$ and $V \leq \beta A_X$. Unit marginal Gaussian tails imply

$$\mathbb{E}[A_X^2 \mathbf{1}_{\{A_X > T\}}] \leq Cm(T^2 + 1)e^{-T^2/2}.$$

Splitting according to $V \leq \beta T$ gives

$$\mathbb{E}V^2 - \mathbb{E}\sum_i f_i z_i^2 \leq 2\beta T \mathbb{E}c_f + C\beta^2 m(T^2 + 1)e^{-T^2/2}. \tag{9.8}$$

Take $T = \sqrt{32L}$. Combining (9.6)–(9.8) and $\beta T \leq CL/m$ gives

$$\mathbb{E}\sum_i f_i z_i^2 \geq \mathbb{E}V_*^2 - \frac{CL^3 J}{m} - C\beta^2 mLe^{-16L}. \tag{9.9}$$

9.4 A quadratic certificate with unequal actual masses

Define

$$Q_i = \mathbb{E}[f_i(XX^T - I)], \qquad T_i = Z_i Z_i^T - \frac{Z^T Z}{m}.$$

Since $\sum_i Q_i = 0$, every test row has squared norm β^2, and $\sum_i p_i = 1$, the numerator is

$$T_f := \sum_i \langle Q_i, T_i\rangle_{\text{F}} = \mathbb{E}\sum_i f_i z_i^2 - \sum_i p_i |Z_i|^2 = \mathbb{E}\sum_i f_i z_i^2 - \beta^2. \tag{9.10}$$

Thus exact balance is not required. The denominator satisfies

$$\sum_i \|T_i\|_{\text{F}}^2 = m\beta^4 - \frac{\text{tr}((ZZ^T)^2)}{m} \leq D_* := \frac{\alpha^4(m-1)(m-2)}{m}, \tag{9.11}$$

because $\text{tr}(ZZ^T) = w$ and $\text{rank}(ZZ^T) \leq m - 1$. Consequently $A_2(f) = \frac{1}{2}\sum_i \|Q_i\|_{\text{F}}^2 \geq (T_f)_+^2/(2D_*)$ by Cauchy's inequality. This remains true if the actual test tensor vanishes.

For the regular reference,

$$T_* := \mathbb{E}V_*^2 - \beta^2 = \alpha^2\{\mathbb{E}[(\max_i Z_i^{\mathrm{iid}})^2] - 1\} > 0, \qquad A_2(g) = \frac{T_*^2}{2D_*}, \qquad T_*/D_* \le Cm.$$

For any $e \ge 0$, $(T_* - e)_+^2 \ge T_*^2 - 2T_*e$; hence (9.9)–(9.11) imply

$$A_2(f) - A_2(g) \ge -CL^3J - CL^2e^{-16L}.$$

The exterior floor (9.3) pays the absolute error: $L^2e^{-16L}/J \le Cm^{-13}L^{5/2} = o(1)$. Therefore

$$A_2(f) - A_2(g) \ge -CL^3J. \tag{9.12}$$

This is an exterior estimate for the localized assignment. It is not a global relative second-chaos inequality near the regular partition.

9.5 The entire signed higher tail

Define $\mathcal{H}_r(f) = \sum_{k\ge3}(-r)^k A_k(f)$, and let g_0 be the regular assignment associated with a closest regular centroid B_0. Its linear routing loss is

$$R_0 = w - \langle B_0, B\rangle = \frac{\Delta + d^2}{2} \ge 0.$$

Let $\varepsilon = \frac{1}{2}\|f - g_0\|_{L^1}$. On each point the total mass sent away from the regular winning label pays at least the top-two gap. Equation (2.11), followed by the elementary bathtub estimate, gives

$$\varepsilon \le \min\{1, C\sqrt{mR_0}\}. \tag{9.13}$$

This holds for fractional assignments too.

Put $\tau^2 = 1/(4L)$. Hypercontractivity and $p_i \le 3/(2m)$ give

$$\|T_\tau f\|_2^2 + \|T_\tau g_0\|_2^2 \le C/m.$$

For the difference, write $e_i = \mathbb{E}|f_i - g_{0,i}| \le C/m$. Since $|f_i - g_{0,i}| \le 1$,

$$\|T_\tau(f_i - g_{0,i})\|_2^2 \le e_i^{2/(1+\tau^2)} \le Ce_i/m.$$

Summation yields $\|T_\tau(f - g_0)\|_2^2 \le C\varepsilon/m$. On degrees at least three, factor the exact signed multiplier through T_τ; the residual operator has norm at most $(r/\tau^2)^3$. Cauchy's inequality therefore gives

$$|\mathcal{H}_r(f) - \mathcal{H}_r(g_0)| \le Cr^3L^3\frac{\sqrt{\varepsilon}}{m}. \tag{9.14}$$

All higher degrees remain in this estimate, with their signs; no finite chaos truncation is substituted.

If $J \ge d^2$, then $R_0 \le J$, and (9.13)–(9.14) and the exterior floor give

$$\frac{|\mathcal{H}_r(f) - \mathcal{H}_r(g_0)|}{J} \le \frac{Cr^3L^3}{m^{3/4}J^{3/4}} \le Cr^3m^{3/2}L^{27/8}.$$

If $J < d^2$, then $R_0 \le d^2$. Use $d = \alpha q$, $J \ge cq/m^3$, and $q \ge c/\sqrt{L}$ to obtain

$$\frac{|\mathcal{H}_r(f) - \mathcal{H}_r(g_0)|}{J} \le Cr^3m^{7/4}L^{13/4}q^{-1/2} \le Cr^3m^{7/4}L^{7/2}.$$

The latter also dominates the first case. Rotation invariance of the regular Hermite weights gives

$$|\mathcal{H}_r(f) - \mathcal{H}_r(g)| \le Cr^3m^{7/4}L^{7/2}J. \tag{9.15}$$

9.6 Completion of the proof

Hermite orthogonality gives exactly

$$S_{-r}(f) - S_{-r}(g) = D + r^2\{A_2(f) - A_2(g)\} + \mathcal{H}_r(f) - \mathcal{H}_r(g).$$

Equations (9.12) and (9.15), with $J = D/r$, therefore imply

$$S_{-r}(f) - S_{-r}(g) \geq D\{1 - CrL^3 - Cr^2m^{7/4}L^{7/2}\} \geq D/2 > 0 \tag{9.16}$$

for sufficiently large m, uniformly in $0 < r \leq 1/(m-1)$. The two endpoint errors are $O(L^3/m)$ and $O(L^{7/2}/m^{1/4})$. Strictness follows from $D = rJ > 0$ in the exterior. This contradicts competitiveness. Consequently every competitive assignment lies inside $d \leq 1/m$, where Analytic interface 2.5 forces the regular assignment. Noncompetitive assignments already have strictly larger noise value. This proves both the inequality and its equality classification in Theorem 1.1, directly for all measurable simplex-valued assignments. No full-noise attainment, purity, stationarity, interface regularity, or limiting optimization argument is required. All thresholds are independent of finite N and of positive r in the voting interval. □

10 Discrete plurality and the fixed-alphabet cut threshold

This section applies the primary transfer results of Isaksson–Mossel [19], with the precise negative-range and single-rule bookkeeping in the endpoint-only transfer note [36]. Its conclusions inherit the proof-candidate status of Theorem 1.1. Write $s_m(\rho) = S_\rho(g)$.

On $[m]^k$ use uniform product measure. The coordinate noise channel is

$$\mathbb{P}\{Y_j = a \mid X_j = b\} = \rho\mathbf{1}_{\{a=b\}} + (1-\rho)/m, \tag{10.1}$$

independently for $j \leq k$. Its allowed range is $-1/(m-1) \leq \rho \leq 1$. The stability of a simplex-valued rule is $\mathbb{E}\langle F(X), F(Y)\rangle$. With any orthonormal product Fourier basis, put

$$\mathrm{Inf}_j^{\leq d_0}(F_i) = \sum_{\substack{\sigma_j \neq 0 \\ |\operatorname{supp}\sigma| \leq d_0}} \widehat{F}_i(\sigma)^2.$$

The ordinary influence is the same sum without the degree restriction.

Corollary 10.1 (Negative Plurality is Stablest)**.** *For every fixed $m \geq m_0$, every $\rho \in [-1/(m-1), 0)$ and every $\varepsilon > 0$, there are $d_0 \geq 1$ and $\tau > 0$, depending only on (m, ρ, ε), with the following property. For every voter count $k \geq 1$ and every $F : [m]^k \to \Delta_m$ satisfying $\mathrm{Inf}_j^{\leq d_0}(F_i) \leq \tau$ for every i, j,*

$$S_\rho(F) \geq s_m(\rho) - \varepsilon = \lim_{\ell\to\infty} S_\rho(\mathrm{Plurality}_{\ell,m}) - \varepsilon.$$

No output-balance, label-equivariance, unanimity, or purity assumption is imposed. The influence parameters are independent of k; no explicit or uniform-in-m influence threshold is claimed.

Proof. Theorem 7.1 of [19], at the fixed parameters in the statement, transfers F to a pure Gaussian assignment h in dimension $k(m-1)$ with the same masses and $|S_\rho(h) - S_\rho(F)| \leq \varepsilon$. This finite dimension is at least $m-1$, so Theorem 1.1 applies. Its absence of a mass constraint is essential here. Lemma 7.3 of [19] identifies the Gaussian regular value with the plurality limit by the joint central limit theorem; Gaussian ties have measure zero, so the finite plurality tie convention does not affect the limit. These two statements give the displayed inequality. □

The stronger ordinary-influence assumption also suffices. So does a bound by τ on the sum of the ordinary influences of all output labels at each coordinate. The negative endpoint of (10.1) is included: a coordinate then changes uniformly to a different label. At $\rho = 0$, the conclusion follows without any influence restriction from $S_0(F) = \|\mathbb{E}F\|^2 \geq 1/m$. There is no implication here for positive noise or for a channel below the negative endpoint.

10.1 The exact Frieze–Jerrum constant

Define

$$\mathfrak{a}_m = \min_{-1/(m-1)\leq\rho\leq 0} \frac{m}{m-1}\frac{1-s_m(\rho)}{1-\rho}. \tag{10.2}$$

This is an exact fixed-m constant. The endpoint characterization of García-Soriano–Miyauchi [37, Theorems 4.8–4.9] gives

$$\mathfrak{a}_m = 1 - s_m(-1/(m-1)). \tag{10.3}$$

That endpoint analysis and rounding-specific tightness are credited prior results. The new Gaussian inequality supplies the separate ingredient for matching hardness against all algorithms through the UGC reduction. The value of the ratio at zero is 1.

For $0 \leq \rho < 1$, purity of g and its nonnegative Hermite weights give

$$s_m(\rho) = \frac{1}{m} + \sum_{j\geq 1} \rho^j A_j(g) \leq \frac{1}{m} + \rho\left(1 - \frac{1}{m}\right).$$

Thus the regular ratio is at least 1 throughout the positive interval. Its usual Frieze–Jerrum infimum over the full channel interval is therefore exactly (10.2). This is the negative-range reduction of [19, Lemma A.4]; no positive-noise simplex optimality is needed.

Corollary 10.2 (Sharp fixed-m MAX-m-CUT threshold under UGC)**.** *For every fixed integer $m \geq m_0$ and every $\varepsilon > 0$, there is a randomized polynomial-time algorithm for nonnegative rational edge-weighted MAX-m-CUT with approximation ratio $\mathfrak{a}_m - \varepsilon$ in expectation, and with high probability after repetition and suitable internal slack. Assuming the Unique Games Conjecture, approximating this problem within $\mathfrak{a}_m + \varepsilon$ is NP-hard. The algorithm and reduction may depend on the fixed m, ε.*

Proof. An edge of the Frieze–Jerrum semidefinite relaxation has vector correlation $\rho \geq -1/(m-1)$ and SDP contribution $(m-1)(1-\rho)/m$. Regular-simplex rounding cuts it with probability $1-s_m(\rho)$. Equation (10.2) and the positive-range estimate above give the edgewise ratio $\mathfrak{a}_m$. Polynomial-precision solution and rounding give arbitrary small slack, as in [19, Appendix A.3, Theorem A.6].

For hardness, the UGC reduction in [19, Appendix A.4], in its negative-correlation form, has completeness arbitrarily close to $(m-1)(1-\rho)/m$ and soundness arbitrarily close to $1 - \Lambda_m^-(\rho)$, where Λ_m^- is the unrestricted Gaussian minimum. Theorem 1.1 identifies it with $s_m(\rho)$ for every negative channel correlation. Dimensions below $m-1$ are covered by cylindrical embedding. Use the rational endpoint $\rho = -1/(m-1)$, where the completeness parameter is one, and choose the invariance and decoding errors sufficiently small in the order given in [36]. The resulting hardness ratio is arbitrarily close to $\mathfrak{a}_m$ from above.

The soundness calculation is genuinely single-rule. Conditional on a verifier vertex v, its independently sampled neighbors define the averaged simplex-valued function $G_v(x) = \mathbb{E}_w[\mathbf{1}_{\{F_w(\sigma_{vw}x)=i\}}]_{i=1}^m$. The rejection probability is exactly $\mathbb{E}_v S_\rho(G_v)$. Corollary 10.1 supplies the low-degree soundness and decoding step for this same function in both slots. No inequality for two independently chosen Gaussian rules is used. □

The approximation statement is unconditional; UGC is needed only for the hardness assertion. We state the primary weighted formulation. The algorithm and hardness reduction may depend on the fixed m and approximation slack. No deterministic derandomization, explicit numerical m_0, or running-time claim uniform in a growing input alphabet is made. The older asymptotic endpoint result [20, Proposition 11.4] does not by itself identify the exact constant (10.2) for every fixed sufficiently large m.

A Proof dependencies, formalization boundary, and provenance

A.1 How to read the supplied proof package

The article gives the statement, the principal analytic reductions, and the complete final comparison. The longer weak Hodge, finite-fan cycle, moving-boundary, and all-amplitude cap calculations remain in the supplied ancillary notes. The PDF alone is therefore not a self-contained replacement for those notes. They are mathematical parts of this submission, not external theorems inferred from successful tests or audit verdicts.

The source archive contains the directory `anc/`, with the following stable layout. Paths in the references are relative to the archive root.

Directory	Mathematical role
`anc/original/`	Preserved analytic source notes, including the stronger cycle-refined estimate, local cap, offset duality, and affine signed-tail proofs.
`anc/integrated/`	Current local, exterior, assembly, and arbitrary-PSD ray reconstructions, with section-level dependencies.
`anc/classical-hodge/`	Covariance concavity, finite-simplex normal strain, established regularity replacement, and endpoint-only discrete transfer.
`anc/independent-hodge/`	Supporting Hodge and covariance-transfer derivations and finite algebra checks.
`anc/independent-hazard/`	Independent reconstruction of the threshold/hazard derivative calculation.
`anc/current-diagnostics/`	Additional reconstruction of the critical mixed estimate and a reproducible, nonrigorous tetrahedral diagnostic.
`anc/lean/`	Pinned Lean sources, theorem inventory, axiom audit, and precise formalization coverage.

The current integrated notes have four distinct roles:

1. [29, Sections 2–4] supplies the initial mass box, the cycle-refined mass envelope, improved localization, and uniform quantifiers. Its use of the mixed offset estimate is explicit.
2. [31, Sections 1–6] supplies the arbitrary-assignment local comparison, including the all-amplitude cap and the retained positive centroid square.
3. [32, Sections 2–9] supplies the hazard derivative, summed-row payment, equal-row test, quadratic certificate, and signed-tail comparison outside the score ball.
4. [30] supplies the finite facet bootstrap and the extension along a ray to every PSD covariance, including singular endpoints.

The branch interface closure note in `anc/integrated/assembly/` records their matching hypotheses. It supplies no new theorem by itself.

A.2 External mathematical inputs

The modern Gaussian maximum comparison is Mulgund's Theorem 2.1 [15]: common thresholds, unit variances, and $K \succeq \mathbf{1}\mathbf{1}^T/m$, including singular covariances. The transformation in Section 9 verifies these exact hypotheses. Neither a general rectangular comparison nor unequal marginal variances are assumed.

The Hodge successor uses Colombo–Figalli–Jhaveri regularity [39] through the reciprocal-coordinate reduction. It replaces the earlier recent-preprint regularity input in that branch. It does not remove the cycle-refined Ehrhard argument from the mass envelope. The classical Gaussian isoperimetric, intrinsic Ehrhard, Brascamp–Lieb/Poincare, and hypercontractive inputs are identified in Section 2.

The discrete transfer is credited to Isaksson–Mossel [19]. The Frieze–Jerrum endpoint characterization is credited to García-Soriano–Miyauchi [37]. Their tightness analysis for that rounding scheme is distinct from the matching UGC-hardness consequence claimed here. Heilman's three-label result [38] is prior work; this manuscript neither reproves it nor claims a new result for $m = 3$.

A.3 What the Lean development establishes

The accompanying Lean project is a partial formalization. Its individual theorems and explicit hypotheses, pinned toolchain, and coverage ledger determine its precise scope. Its concrete Gaussian components prove hyperplane nullness for product Gaussians, almost-everywhere unique winning linear scores, exponential and first-moment integrability, and integrability of finite linear scores and their maximum. These facts support fractional routing rigidity from a zero expected maximum gap. Finite centroid-pairing and covariance edge-strain identities are also formalized. The scalar analysis includes shifted-logarithm decay, the exact endpoint power comparison, and a single threshold uniform over the voting interval for fixed supplied constants. A conditional local/exterior assembly consumes the analytic estimates explicitly. Gaussian transport, hypercontractivity, facet regularity, and the source papers' transfer theorems are not supplied by these component proofs. A checked implication from analytic estimates does not prove those estimates.

In particular, this manuscript does not claim a Lean proof of Theorem 1.1, its analytic interfaces, the uniformity of their constants in dimension and noise, or the UGC reduction. Scalar eventual threshold statements concern fixed supplied constants; they do not establish the Gaussian estimates or the uniformity of their constants. The formal development is a foundation for further work, with its unformalized obligations stated explicitly. Source checksums certify file identity. Compiling this PDF certifies typesetting. Neither operation certifies the analysis.

A.4 Research status and assistance

The author used AI assistance to develop and organize the arguments, perform separate internal reconstructions, prepare the manuscript, and implement checking code. The internal reviewers were AI-agent runs; they are not external experts and their verdicts are not assumed in any mathematical inference. External expert review and exhaustive priority clearance remain pending. No cited author is claimed to endorse this manuscript.

The public version history and immutable release records are available at `https://github.com/infinityscroll/jmlr-research-notes`. They establish a record of the released artifact, not mathematical correctness or exclusive research priority. Earlier frozen

manuscripts and their source notes are preserved alongside this integrated edition. The existential large-alphabet threshold, full negative voting interval, and fixed-m computational interpretation are the claims of this edition. The wider-correlation supplements are outside its scope.

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